

ExitCheck Project Manual

SYSTEM SETUP & OPERATOR REFERENCE GUIDE

1. Project Introduction

ExitCheck is a smart home orchestration utility that aggregates contextual information into a mobile-first dashboard (HUD). Triggered by tapping an NFC tag placed by your primary exit, the system gathers local sensor data, driving conditions, transit paths, calendar invites, and weather updates to prepare you for your departure in under 10 seconds.

The dashboard functions in a browser sandbox environment on the user's terminal and is designed for an active, glanceable interface before stepping out the door.

2. Detailed Setup Steps

Step 2.1: Programming the NFC Sticker

The physical trigger for ExitCheck is a passive NFC tag (NTAG213/NTAG215). Follow these steps to map it:

1. Download `NFC Tools` (or any write tool) on your Android or iOS device.
2. Navigate to **Write** -> **Add a Record** -> **URL/URI**.
3. Input the following URL replacing the user ID if custom profiles are needed:
`http://localhost:3000/exit?tag_id=door_main&user_id=usr_123`
4. Hold your smartphone against the tag to write. Stick the tag next to your front door handle or keys tray.

Step 2.2: Setup Key Addresses

Go to the Settings page in the dashboard to set up your standard commuting addresses. Enter the full street address for your Home and Work. These will be parsed and formatted by Nominatim to resolve GPS positions.

IMPORTANT: BROWSER GEOLOCATION

Ensure that Safari or Chrome has permission to read location services on `localhost:3000` . If location is denied, the HUD falls back to the geocoded position of your saved Home address.

3. System Architecture

ExitCheck utilizes a Next.js framework backed by SQLite and Prisma, communicating concurrently with external APIs.

Component	Integration Provider	Purpose
Geocoding	Nominatim (OpenStreetMap)	Resolves textual addresses to coordinates without API keys.
Commute Calculations	OSRM (Open Source Routing Machine)	Calculates accurate live driving paths and distances (in miles).
Weather Calculations	Open-Meteo API	Free weather details parsed from WMO codes.
Calendar Integration	Google Calendar API	OAuth 2.0 connection to read the next active calendar invite.
Database Engine	Prisma & SQLite	Manages user address profiles, gear items, and OAuth session tokens.
AI Synthesis	Google Gemini 1.5 Flash	Orchestrates context maps to return a structured JSON response.

4. Core API Documentation

HUD Aggregator API

Endpoint: `POST /api/exit`

Payload schema:

```
{
  "tag_id": "door_main",
  "user_id": "cmskb90a2000siv5kzgw24il",
  "latitude": 40.7321,
  "longitude": -74.0606,
  "batteryLevel": 0.85,
  "isCharging": false
}
```

Action Button Dispatcher API

Endpoint: `POST /api/actions`

Dispatches quick IoT webhooks to control smart home systems. Schema:

```
{
  "actionId": "garage_door_close" // Or "tesla_climate_start"
}
```

5. Troubleshooting

Next.js port conflicts: If the port is in use, check background processes with `lsof -i :3000` and terminate frozen processes using `kill -9 <PID>` before restarting `npm run dev`.

AI Rate-limits: Gemini free tier has a 20 request/day quota. ExitCheck includes a graceful fallback cache. If your quota is reached, real OSRM travel maps still operate, but static placeholders are shown for AI alerts.